

XAI: Model-agnostic Methods

Partial Dependence Plots

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Evaluation, Deployment and Monitorisation of Models

1 Introduction

This report uses Partial Dependence Plots (PDPs) on two prediction tasks: predicting daily bike rental counts from the Capital Bikeshare dataset [1], and predicting house sale prices from the King County housing dataset [2]. In both cases a random forest is used as the model, and PDPs are used to show how individual features affect the predictions.

Random forests are flexible and accurate, but hard to interpret on their own. PDPs help with this by fixing all other features and showing how the prediction changes as one feature varies. This makes it possible to see the relationship the model has learned for each feature separately. The report is split into three parts: one-dimensional PDPs for the bike rental model (Section 2), a two-dimensional PDP for temperature and humidity together (Section 3), and one-dimensional PDPs for house prices (Section 4).

2 One-Dimensional PDPs: Bike Rentals

2.1 Setup

The `day.csv` dataset [1] was used without further preprocessing beyond computing `days_since_2011` as the number of days elapsed since 1 January 2011. A random forest was fitted using the `randomForest` package [3] with 500 trees and a fixed seed of 42, predicting the daily rental count `cnt` from four features: `days_since_2011`, `temp`, `hum`, and `windspeed`. Partial dependence was computed with the `pdp` package [4] and the plots were produced with `ggplot2` [5]. Rug marks at the bottom of each panel show the actual distribution of observations along each axis.

2.2 Results

Figures 1–4 show the one-dimensional PDPs for each feature.

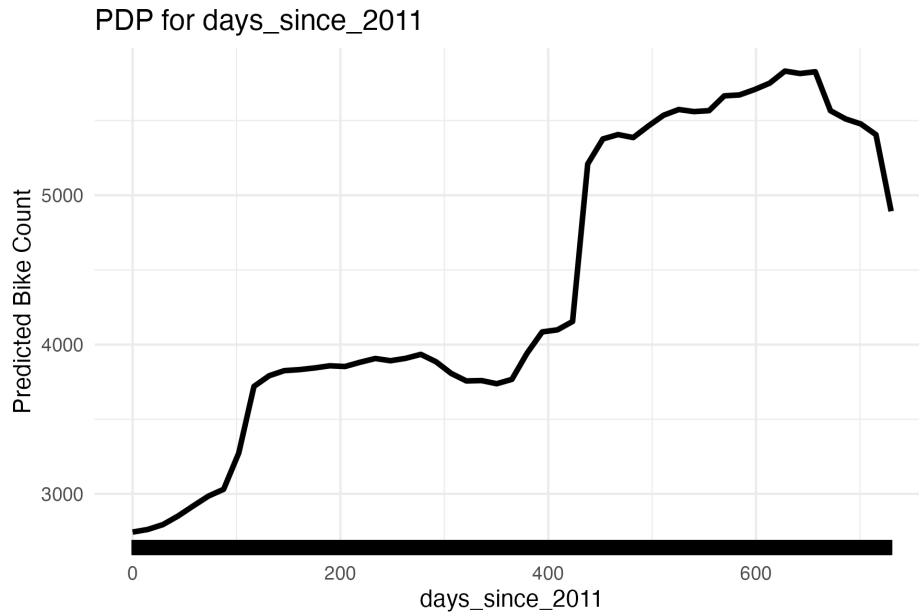


Fig. 1: PDP for `days_since_2011`. Predicted bike count increases approximately linearly across the observation period, reflecting steady service growth from 2011 to 2012.

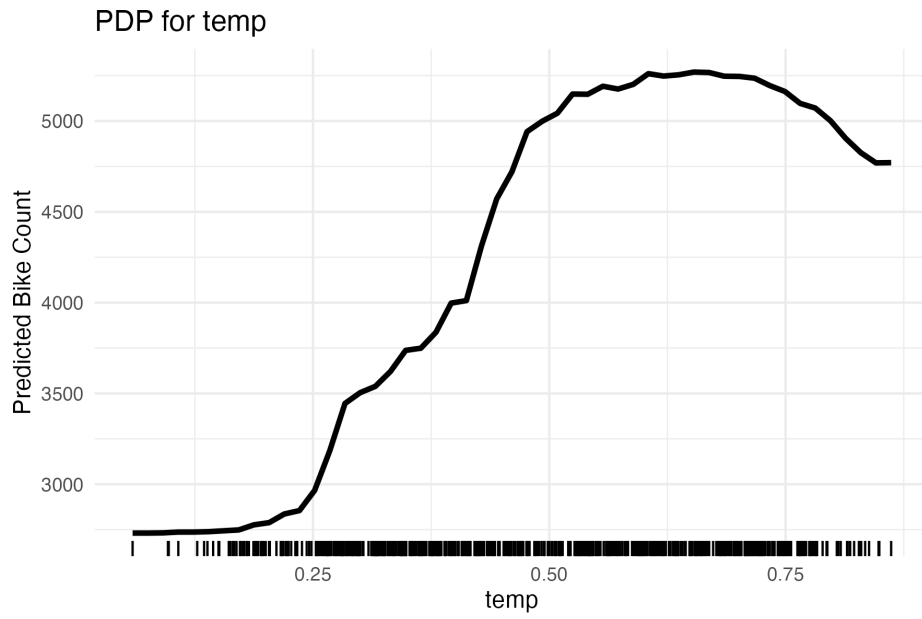


Fig. 2: PDP for `temp` (normalised). Rentals increase sharply with temperature up to moderate warmth, then plateau and show a slight decline at extreme heat.

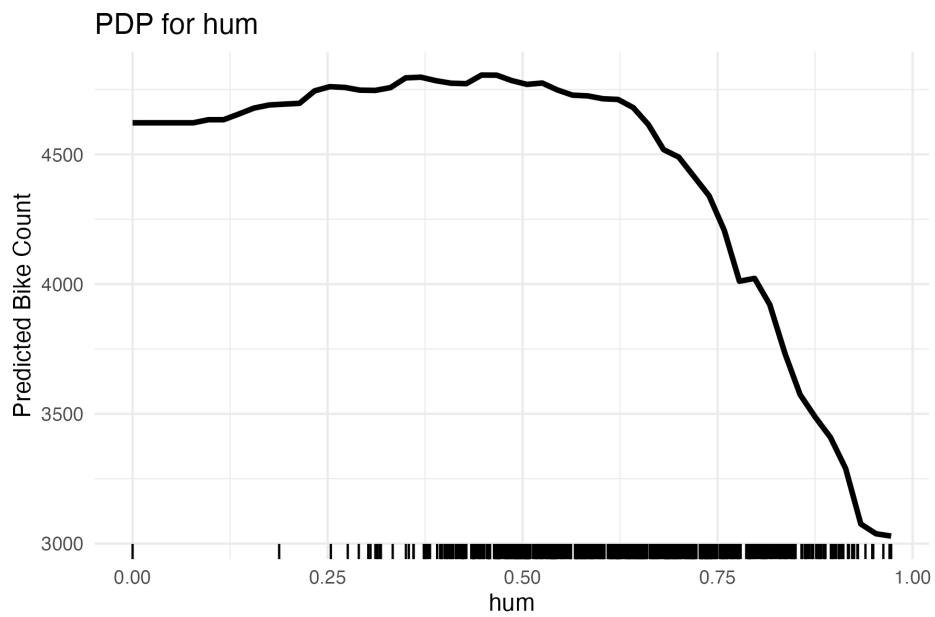


Fig. 3: PDP for `hum` (normalised). Predicted count is relatively stable at low to moderate humidity, then falls off noticeably above roughly 0.75 normalised humidity.

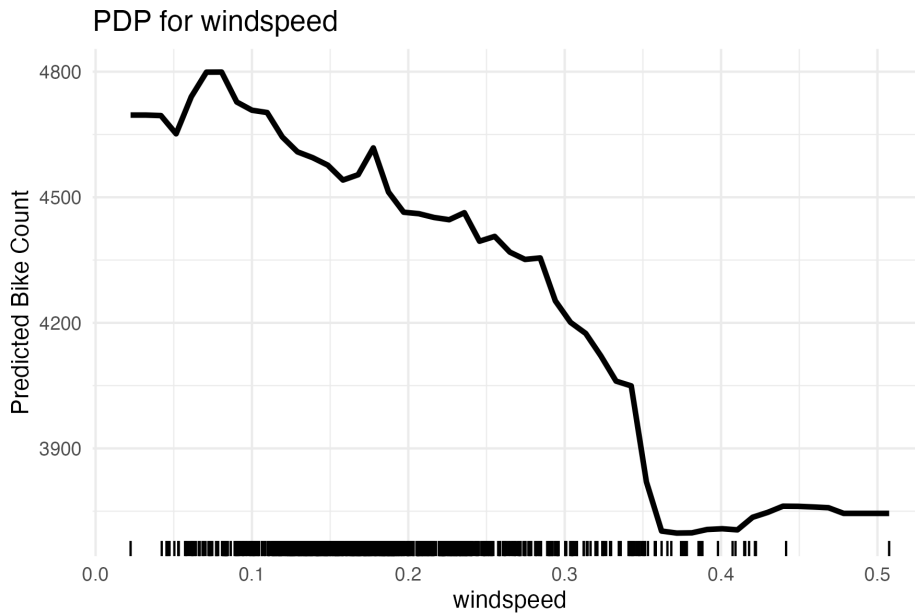


Fig. 4: PDP for `windspeed` (normalised). Wind speed has a consistently negative effect on predicted rentals, with the steepest drop at low-to-moderate speeds and a flatter tail at high values.

2.3 Interpretation

Days since 2011. Figure 1 shows a clear upward trend from around 3 500 predicted rentals at the start of 2011 to nearly 5 500 by the end of 2012. The increase is steady throughout, with no sudden jumps at any point in time. This matches what the linear model in Exercise 1 found, where this feature had the strongest effect. Two small dips appear around days 350 and 650, which correspond to late autumn and early winter of 2011 and 2012. This is likely because factors like shorter daylight hours and holiday travel patterns are not included as separate features, so the model uses the time variable as a rough stand-in for them. The main takeaway is that demand grew steadily over the period, and any forecasting model should account for both the overall upward trend and a small seasonal dip in late autumn.

Temperature. Figure 2 shows that rentals increase as temperature rises, up to around 0.6–0.7 on the normalised scale (roughly 15–25 °C). Beyond that point the curve flattens and dips slightly, meaning that very hot days see fewer rentals than warm ones. This makes sense: people cycle most when conditions are comfortable and avoid both cold and extreme heat. A linear model would miss this pattern, since it can only fit a straight line through the data.

Humidity. Figure 3 shows that humidity has almost no effect on predicted rentals at low to moderate levels. Once normalised humidity goes above about 0.75, however, predicted rentals drop sharply. The effect is not gradual - it stays flat and then falls off quickly past a certain point. A linear model would miss this, since it would spread the negative effect evenly across all humidity levels instead of concentrating it at the high end.

Wind speed. Figure 4 shows that wind consistently reduces predicted bike rentals. The drop is sharpest at low-to-moderate wind speeds-where even a small increase causes a noticeable fall in predicted count-and the curve levels off at high values. This makes intuitive sense: on an already very windy day, fewer people cycle regardless, so additional wind has less room to make things worse. Wind speed never has a positive effect on rentals at any observed value.

3 Two-Dimensional PDP: Temperature and Humidity

3.1 Setup

Using the same random forest from Section 2, a 2D PDP was computed for `temp` and `hum` together. The result was plotted as a heatmap using `geom_tile()` in `ggplot2`, with white contour lines added to mark prediction levels. Density plots showing the distribution of each feature were placed above and to the right of the heatmap using `cowplot` [6].

3.2 Results

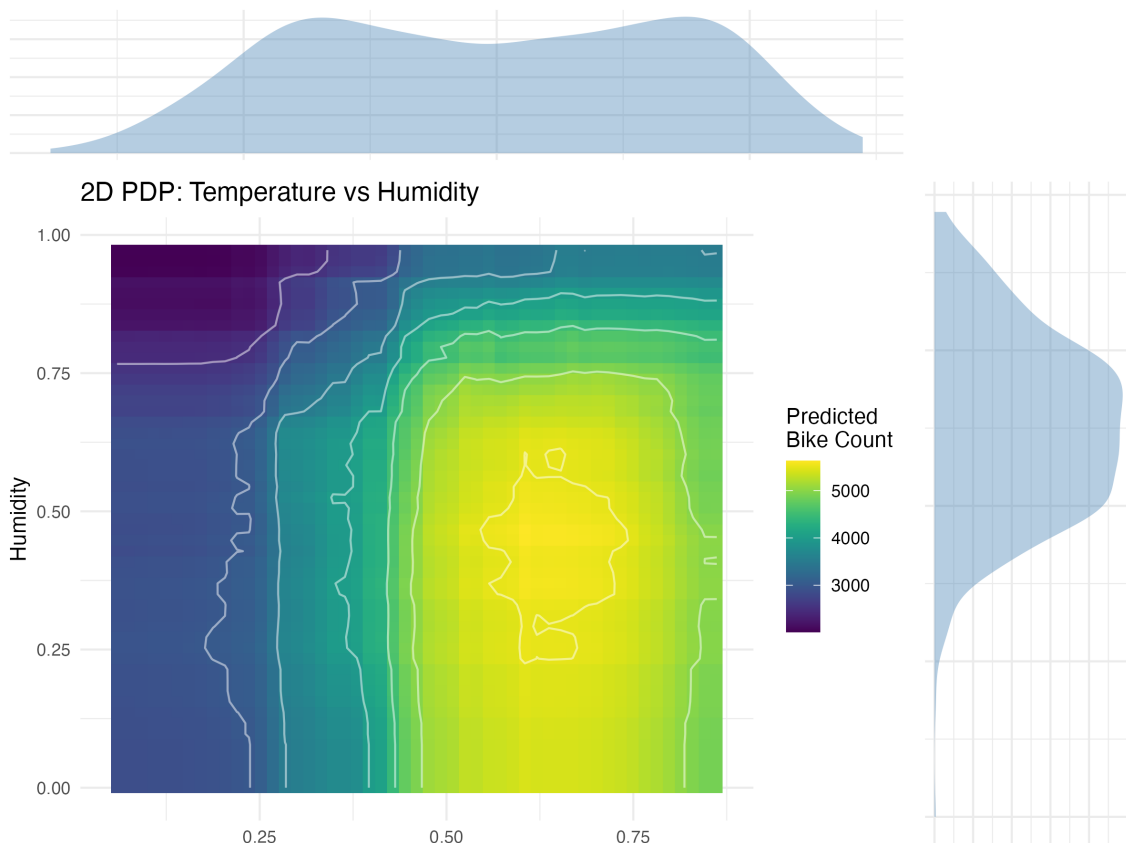


Fig. 5: 2D PDP for `temp` and `hum`. The colour encodes predicted bike count (yellow = high, purple = low). Marginal density distributions are shown above (temperature) and to the right (humidity).

3.3 Interpretation

Figure 5 shows how temperature and humidity together affect predicted rentals. The highest predicted counts (yellow) are in the centre-right of the heatmap, where temperature is moderate-to-high and humidity is low-to-moderate. This matches the 1D temperature PDP, which showed a slight drop at very high temperatures: the top-right corner of the heatmap (high temperature, high humidity) is teal rather than yellow, meaning that a hot and humid day does not produce the highest rentals. The lowest counts (dark purple and dark blue) are on the left side, where cold temperatures keep rentals low regardless of humidity.

Two things stand out that the individual PDPs cannot show. First, humidity barely affects predictions on cold days: the colours on the left side of the heatmap change very little from bottom to top. On warm days, however, moving from low to high humidity clearly shifts the colour from yellow to green, meaning fewer predicted rentals. In other words, humidity only matters when it is already warm - a pattern that two separate 1D plots would not reveal. Second, a wide band of moderate temperatures in the middle of the heatmap produces similar predictions across many humidity levels, forming a broad area of medium demand.

The density plots on the edges show that most days in the dataset fall in the mid-temperature range and below 0.8 humidity. Predictions in the corners of the heatmap - very cold, very hot, or very humid days - are based on fewer observations and are less reliable.

4 PDPs for House Price Prediction

4.1 Setup

The King County housing dataset [2] contains over 21 000 records. To keep computation manageable, a random sample of 5 000 rows was drawn with `set.seed(42)`. A random forest with 300 trees was

then fitted to predict `price` from six features: `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, and `yr_built`. PDPs were computed and plotted for four of these features: `bedrooms`, `bathrooms`, `sqft_living`, and `floors`. Rug marks show where most of the data is concentrated along each axis.

4.2 Results

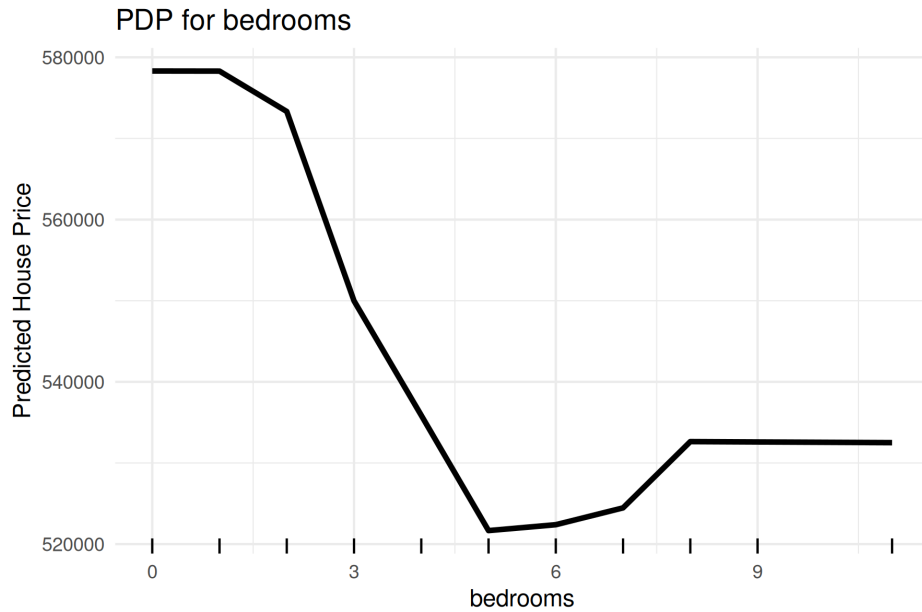


Fig. 6: PDP for `bedrooms`. Predicted price is highest at 0–2 bedrooms and falls sharply from 3 to 5 bedrooms, before partially recovering and plateauing at higher counts.

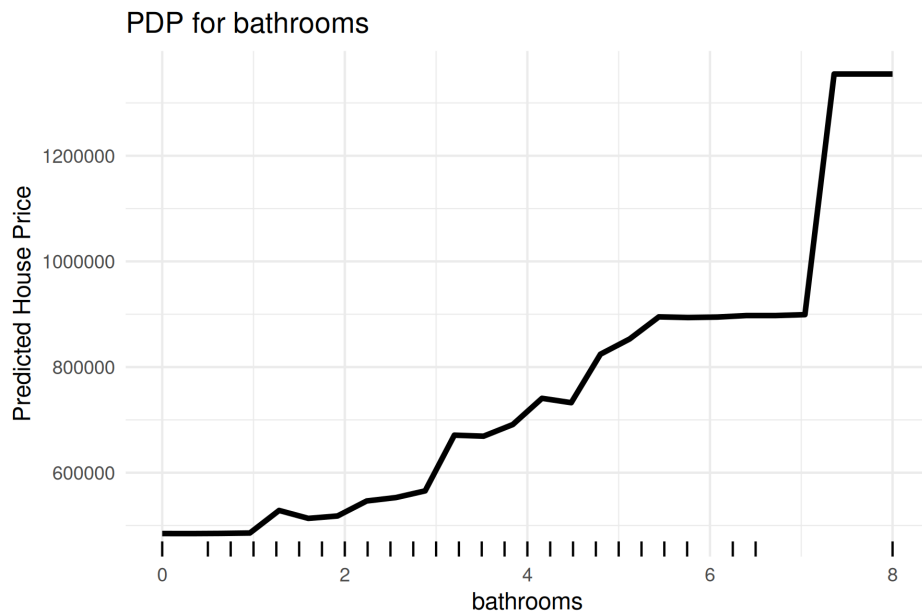


Fig. 7: PDP for `bathrooms`. Predicted price rises overall from around \$460 000 to over \$1.3 million, with a large jump around 2.5–3 bathrooms and another steep rise beyond 7.

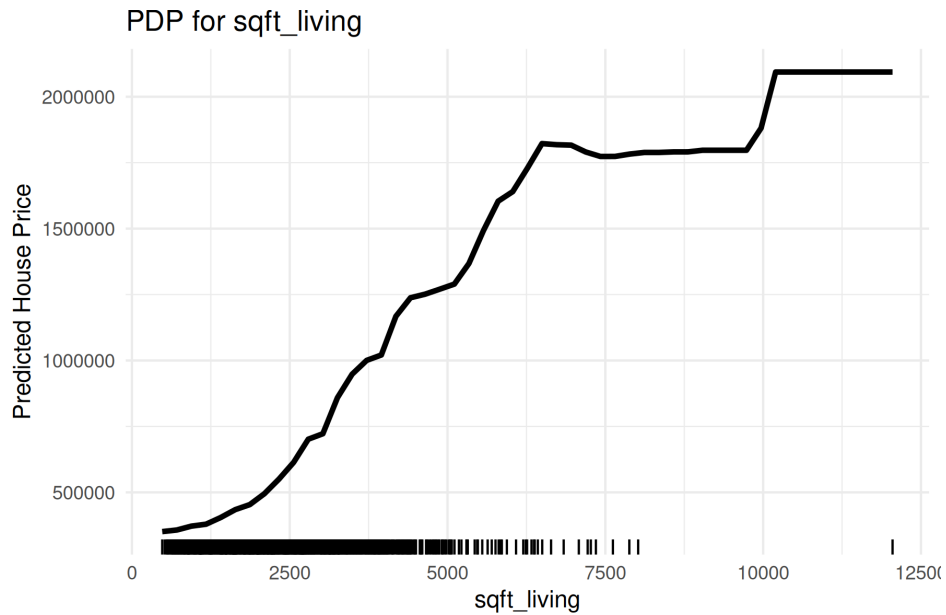


Fig. 8: PDP for `sqft_living`. Predicted price rises from roughly \$320 000 to over \$2 million, with the steepest climb between 1 000 and 5 000 sq. ft., a plateau around 5 000–7 500 sq. ft., and a further rise at very large sizes.

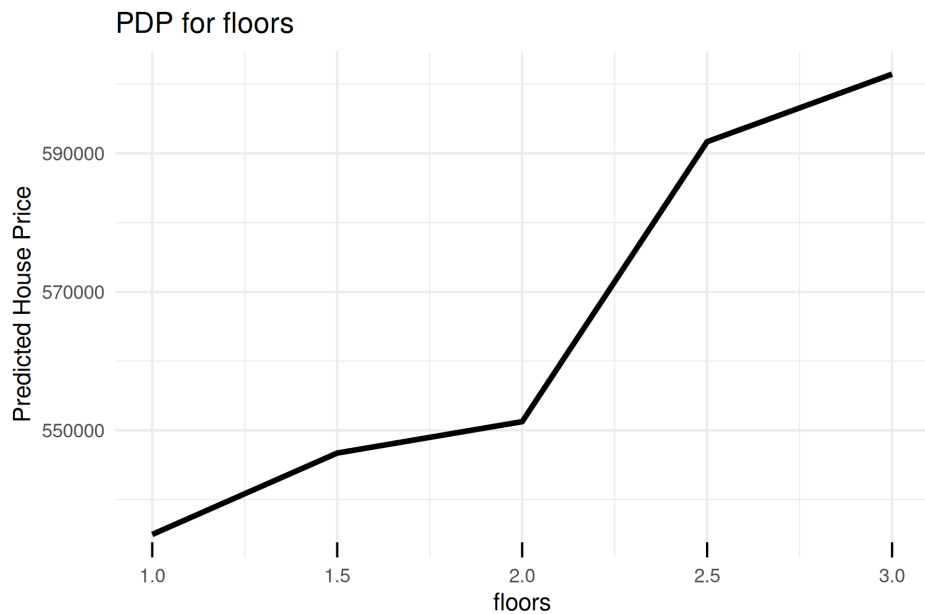


Fig. 9: PDP for `floors`. Predicted price increases monotonically from 1 to 3 floors, with the sharpest jump occurring between 2 and 2.5 floors.

4.3 Interpretation

Bedrooms. Figure 6 shows a surprising result: predicted price is highest at zero to two bedrooms (around \$578 000) and drops sharply as the number of bedrooms grows, reaching a low of about \$522 000 at five to six bedrooms before rising slightly again. This happens because the other features - especially `sqft_living` - are held fixed. If the total living area stays the same, adding more bedrooms just means each room gets smaller, which buyers value less. The small recovery beyond eight bedrooms is based on very few observations, so it should not be read into too much.

Bathrooms. Figure 7 shows that more bathrooms generally means a higher predicted price, from around \$460 000 at zero bathrooms to over \$1.3 million beyond seven. The curve is mostly flat from zero to

two bathrooms, then rises noticeably around 2.5–3, continues upward through five, and levels off before jumping steeply beyond seven. That final spike comes from very few data points and should be treated carefully. In general, more bathrooms signal a larger or better-quality home.

Living area (`sqft_living`). Figure 8 shows the largest price range of any feature, going from about \$320 000 for the smallest homes to over \$2 million at the top. Prices rise most quickly between 1 000 and 5 000 sq. ft., where most of the data sits. Above 5 000 sq. ft. the curve flattens to around \$1.75 million, then climbs again beyond 7 500 sq. ft. before levelling off near \$2.1 million. Living area has the clearest and strongest effect on price of all four features.

Floors. Figure 9 shows that predicted price rises steadily from one floor (around \$525 000) to three floors (around \$598 000), with the biggest jump between two and 2.5 floors. There is no drop at higher values - more floors consistently leads to a higher predicted price. This likely reflects that homes with more floors tend to have more total space or are found in more desirable areas.

Summary. Of the four features, `sqft_living` has the strongest influence on price, spanning a range from \$320 000 to over \$2 million. Bathrooms also push prices up, though the effect is uneven. Bedrooms go against expectations - holding size fixed, more bedrooms actually lowers the predicted price. Floors show a steady positive effect but a smaller overall impact. Together, these PDPs reveal patterns that a simple linear model or a single importance score would not show clearly.

5 Conclusions

This exercise showed how Partial Dependence Plots can make a random forest easier to understand. For bike rentals, the PDPs confirmed that temperature is the most important factor, but also revealed patterns that a linear model would miss: rentals peak at comfortable temperatures and drop in extreme heat, humidity only discourages cycling above a certain threshold, and wind reduces demand most strongly at low-to-moderate speeds. The 2D PDP added an extra insight by showing that temperature and humidity interact - high humidity matters much more on warm days than on cold ones.

For house prices, the results were sometimes surprising. Living area had the strongest and clearest effect on price. Bathrooms showed a consistent positive trend. Bedrooms, however, had a negative effect once other size features were held fixed, because more bedrooms in the same space means smaller rooms. Floors increased price steadily from one to three storeys with no decline. Overall, PDPs proved useful for turning a complex model into a set of clear, readable relationships - which is exactly what explainable AI aims to do.

References

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